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Sonar-Based Underwater SLAM with 3D Gaussian Splatting Mapping

Saxon Shang supervised by Dr. Sen Wang
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Why Underwater Sonar SLAM Is Hard

Can a renderable sonar map provide online correction factors?

1. Problem setting

Underwater robots cannot use GNSS. IMU/DVL odometry therefore drifts over time.

2. Sensor limitation

Sonar works underwater, but the image evidence is sparse, ambiguous, and viewpoint-dependent.

3. Sonar-based SLAM limitation

Geometric matching becomes fragile under sonar noise and repeated acoustic structure.

4. Key insight from 3DGS

A learned Gaussian scene can predict the sonar view from a candidate pose.

5. Gap

3DGS is rarely used as an online sonar measurement model inside SLAM.

6. Project idea

Use render consistency between live sonar and 3D Gaussian Splatting prediction as a SLAM constraint.

Research Gap: Renderable Sonar Maps Inside SLAM

What the literature gives, and what this project adds

Sonar SLAM

- Uses sonar features, occupancy maps, or image factors to reduce drift.
- Still depends on difficult acoustic data association.
- Examples: Westman, Li, McConnell, Aqua-SLAM.

3DGS SLAM

- Uses Gaussian maps for dense tracking and reconstruction.
- Mostly assumes camera or RGB-D observations.
- Examples: Gaussian-SLAM, GS-SLAM, RTG-SLAM, SemGauss-SLAM.

Underwater / sonar GS

- Learns renderable underwater or sonar-like scene models.
- Mainly evaluated as reconstruction or rendering.
- Examples: NAS-GS, Aqua-Splat, SeaSplat, WaterSplatting, UW-GS.

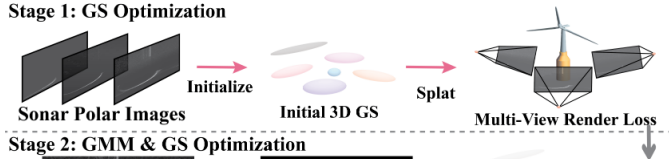
Missing bridge

Use a trained sonar Gaussian map as an online measurement model: candidate Pose3 $\rightarrow$ NAS-GS render $\rightarrow$ live-sonar consistency check $\rightarrow$ gated Pose3 factor in iSAM2.

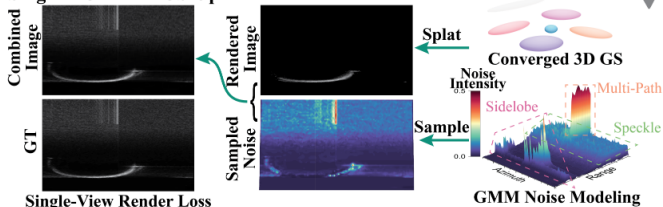
NAS-GS: Noise-Aware Sonar Gaussian Splatting

A renderable sonar measurement model reference

Stage 1: GS Optimization



Stage 2: GMM & GS Optimization



NAS-GS training pipeline, adapted from Xu et al.

What NAS-GS does

- Learns a 3D Gaussian sonar scene from polar sonar images.
- Optimises rendered structure together with acoustic noise appearance.
- Produces a differentiable sonar renderer for new viewpoints.

How I use it

- Freeze the trained map during replay.
- Render candidate Pose3 hypotheses online.
- Insert a factor only when live sonar and render agree.

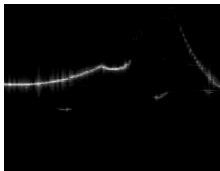
SLAM Pose graph optimization with Pose3 constraints and iSAM2 backend

NAS-GS predicts observations; iSAM2 consumes accepted constraints

Online loop: render from the map, compare with the live frame, then add a graph factor only when the evidence is strong.

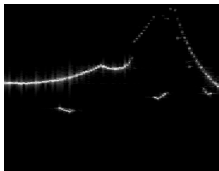
Renderable sonar map

NAS-GS predicts what the sonar should observe from a proposed Pose3 camera pose.



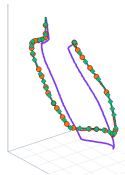
Online evidence

A live sonar frame is compared against rendered candidates around the current drifted pose.



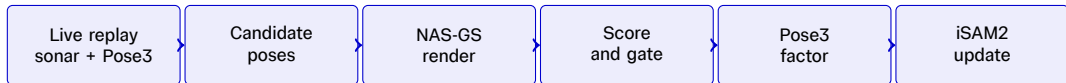
Graph correction

If render similarity and gates pass, the recovered pose is inserted as a Pose3 factor.



Online-render Registration Backbone

Same online loop for same-route correction and revisit correction



Local single

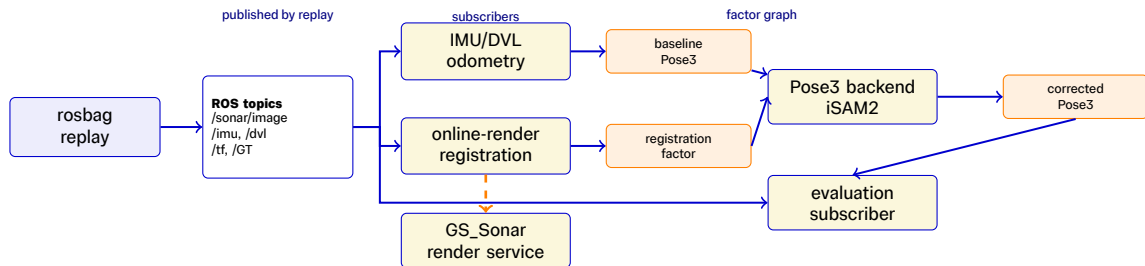
Compact x, y, z, ψ search around the live Pose3 baseline. Main same-route policy for Pass A and off-shore replay.

Route-progress revisit

Sequence render search guided by route progress and drift-prior checks. Used for revisit cases such as Pass B.

Gating is separate from the mode. Relaxed gating accepts useful factors earlier, while score and structure checks still prevent weak render matches from entering the graph.

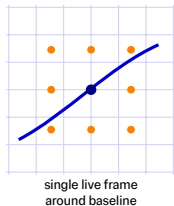
Algorithm Step 1: Live Replay System



The replay is deliberately online. Nodes subscribe to ROS topics in time order, the registration node queries NAS-GS only at candidate poses, and iSAM2 receives only gated Pose3 factors.

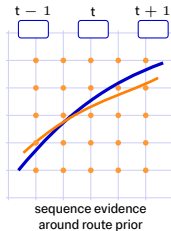
Algorithm Step 2: Candidate Pose And NAS-GS Render

Local single-frame search



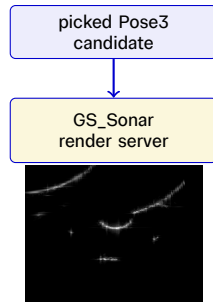
Compact x, y, z, ψ grid for same-route correction.

Route-progress sequence search



The orange line is a smooth candidate correction sequence. Wider candidates are allowed, but neighbouring frames must agree.

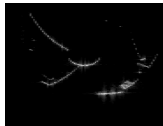
Render query



The output is a rendered sonar image for the proposed robot pose.

Algorithm Step 3: Score And Gate

Accepted pair



live sonar



rendered sonar

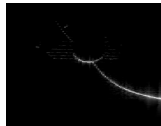
Frame 24 selects rendered frame 24 and passes the score gate.

score gate	$s = 0.926 \geq 0.08$, pass
pose gate	shift = 0.048 m, pass
replay evidence	published as a Pose3 registration factor

Score calculation

Intensity normalisation, NCC-style similarity, residual penalty, and bright-return overlap are combined into one render score. Relaxed threshold $s_{\min} = 0.08$.

Rejected pair



live sonar



rendered sonar

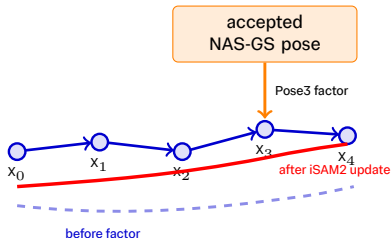
Frame 228 selects rendered frame 216 but falls just below threshold.

score gate	$s = 0.0799 < 0.08$, fail
edge check	best candidate on search edge
replay evidence	one of 6 score-below-threshold events

Gates before insertion

Render-score, structure-support, pose-jump, and temporal-consistency gates must pass before publishing a Pose3 factor to iSAM2.

Algorithm Step 4: Pose3 Factor And iSAM2 Update



Before

The graph contains IMU/DVL odometry factors, so drift accumulates along the chain.

Insertion

The accepted candidate pose and covariance are published as a Pose3 registration factor.

After

iSAM2 optimises the graph, pulling nearby states toward the render-consistent pose.

Algorithm Step 5: Route-progress Revisit Priors



Route prior

$$s_t = \frac{t}{N_B - 1}, \quad i_t = \text{round}(s_t(N_A - 1))$$

Variables: t is the current revisit frame, N_A , N_B are the map and revisit route lengths, s_t is normalised progress, and i_t is the selected Pass A map index.

- This avoids nearest-pose lookup when route direction changes.
- It gives a map window, not the final factor pose.

Drift prior

$$T_{B,t}^{\text{cand}} = T_{B,t}^{\text{odom}} \oplus c_t, \quad c_t \in \mathcal{E}_{\text{drift}}(s_t)$$

Variables: $T_{B,t}^{\text{odom}}$ is the live drifted Pose3, c_t is a sampled correction in x, y, z, ψ , and $\mathcal{E}_{\text{drift}}(s_t)$ is the progress-dependent drift envelope.

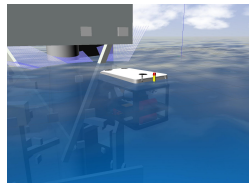
- It is learned from Pass A baseline-to-GT errors.
- It guides search size, not the final accepted pose.

Simulation Platform: DAVE In Docker

Repeatable sonar, IMU, DVL, and ground-truth recording

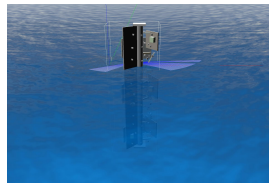
Docker setting and robot model

- CUDA-enabled Docker container.
- DAVE with ROS Noetic and Gazebo.
- BlueROV2 vehicle model.
- Sensors recorded: sonar, IMU, DVL, odometry, and pose ground truth.



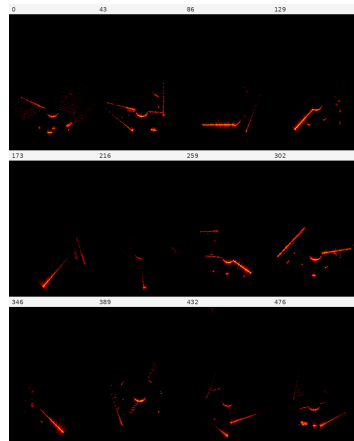
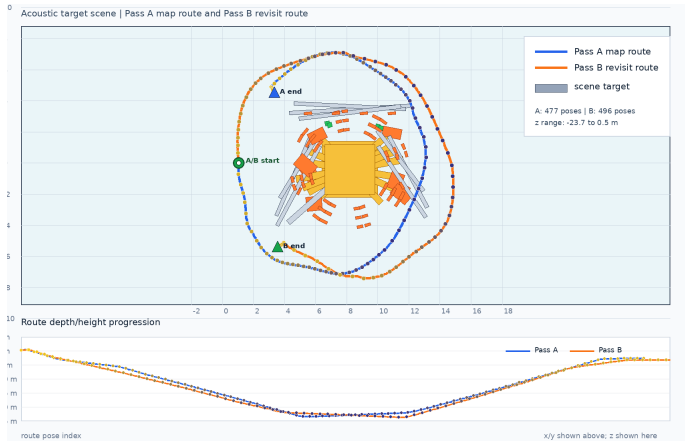
Custom scene description

- A fixed underwater target structure creates repeatable sonar returns.
- Pass A, Pass A revisit, and Pass B use controlled waypoint routes.
- Sonar field of view, range, beam pattern, and noise are explicitly set.



Simulation Path Recording

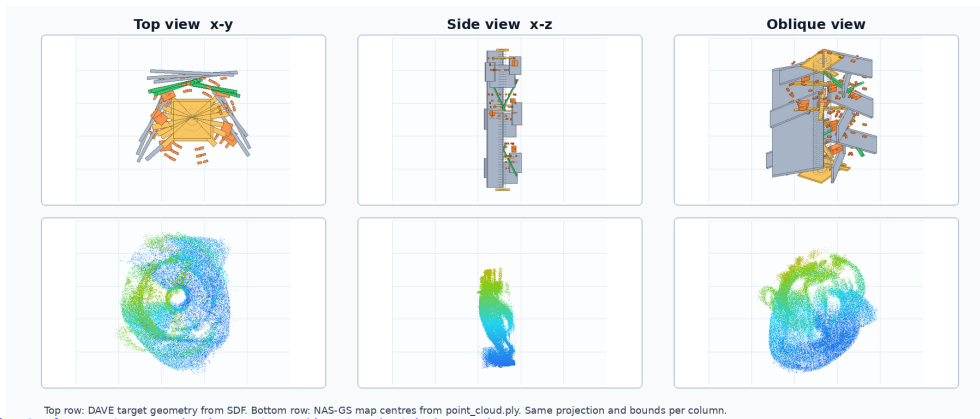
Pass A trains the map; Pass B reuses the start point but follows a hard revisit path



From Simulated Geometry To NAS-GS Map

The learned map

The two rows use the same top, side, and oblique projections. They differ because the DAVE row shows the full simulated target geometry, while the NAS-GS row shows Gaussian centres learned only from sonar-visible acoustic returns.



Experimental Design

Recorded data, map training, and replay policy

Case	Images	Sonar setting	NAS-GS map	Replay policy
Pass A	477	90°, 20 m	Pass A 900k/i1000, PSNR 35.77 dB	Relaxed local single
Pass A revisit	431	90°, 20 m	Pass A 900k/i1000, PSNR 35.77 dB	Relaxed local; route-progress check
Pass B revisit	496	90°, 20 m	Pass A 900k/i1000, PSNR 35.77 dB	Relaxed route-progress sequence
Offshore	1139	120°, 12 m	Offshore 750k/i800, BP PSNR 33.68 dB	Relaxed local single

Baseline setting

IMU/DVL Pose3 replay with NAS-GS factors disabled. Noise is injected as DVL scale 1.10, body bias (0.030, -0.025, 0.0) m/s, yaw drift 8.0×10^{-4} rad/s, and 2.5 s DVL dropout every 35 s after 45 s.

Relaxed gating

Minimum render score 0.08, maximum accepted correction 5 m in x, y and 8 m in z, followed by render-validity and factor-consistency checks.

Pass A Policy Selection

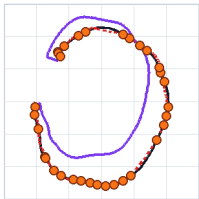
Rendered BP library with runtime no-BP was the most reliable local policy

Pass A online-render with BP and no-BP rendering

Same relaxed local single policy with different NAS-GS render settings

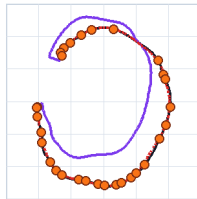
BP90x20 rendering

36 factors XY RMSE 2.741 to 0.733 m mean |z| 12.719 to 0.901 m



no-BP rendering

31 factors XY RMSE 2.741 to 0.612 m mean |z| 12.719 to 0.467 m



— GT sonar mount

— baseline

— corrected

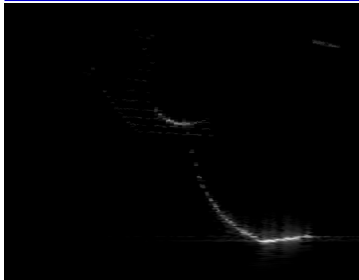
● accepted NAS-GS factor

Pass A Sonar Evidence

One recorded frame and the corresponding BP/no-BP rendered libraries

Recorded Pass A

Live sonar frame from the DAVE route.



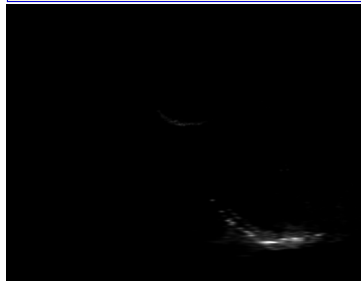
Rendered BP library

Reference image from the trained NAS-GS map.



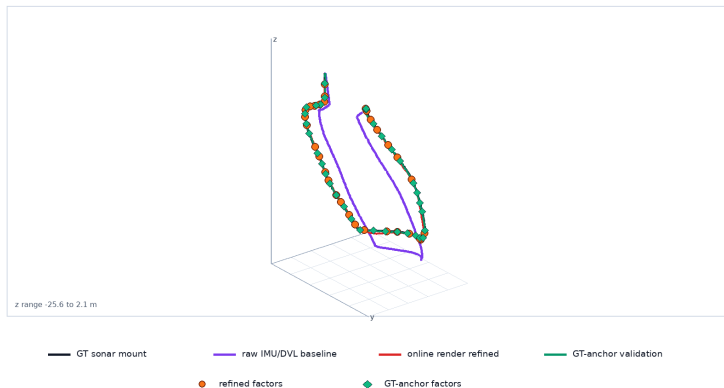
Rendered no-BP library

Alternative rendered appearance for policy testing.



Pass A Result

Pass A 3D z-drift diagnostic with map-pose registration



Metrics: 30 factors. XY RMSE $2.741 \rightarrow 0.570$ m; XYZ RMSE $15.078 \rightarrow 0.937$ m; mean $|z|$ $12.719 \rightarrow 0.569$ m.

Final error: XY 0.305 m; $|z| = 0.069$ m.

BP versus no-BP on Offshore Data

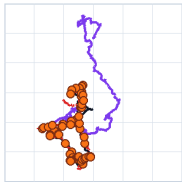
The better rendering mode depends on the recorded sonar appearance

Offshore online-render with BP and no-BP rendering

Same relaxed local single policy on real sonar with different NAS-GS render settings

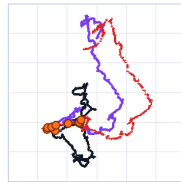
BP rendering

65 factors XY RMSE 11.398 to 1.107 m mean |z| 6.129 to 0.693 m



no-BP rendering

12 factors XY RMSE 11.398 to 12.550 m mean |z| 6.129 to 1.261 m



— GT sonar mount

— baseline

— corrected

● accepted NAS-GS factor

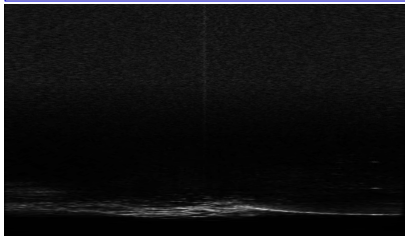
Offshore data contains stronger real-image artefacts, so BP and no-BP rendering are less interchangeable than in the DAVE Pass A replay.

Offshore Sonar Evidence

The real replay has stronger appearance mismatch than the DAVE route

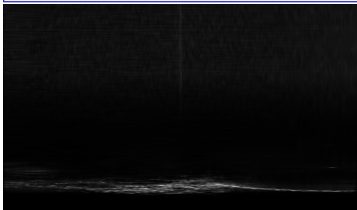
Recorded offshore

Live sonar from the replay.



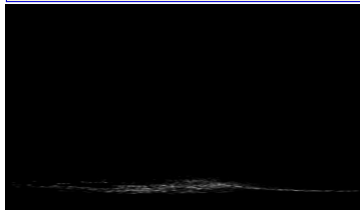
Rendered BP library

NAS-GS render with the offshore BP library.



Rendered no-BP library

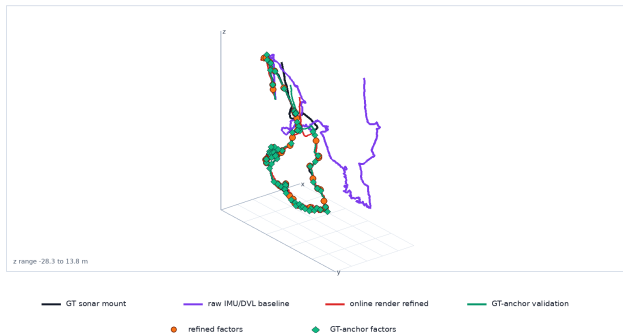
No-BP render used for appearance comparison.



Offshore Validation

The method transfers, but the real-scene case exposes z and appearance limits

Offshore 3D pose graph with live online-render correction



GT-anchor curve is the archived upper-bound diagnostic. Baseline and online-render are current live replay results.

Metrics: 50 factors. XY RMSE 11.043 $\rightarrow$ 1.407 m; XYZ RMSE 12.795 $\rightarrow$ 2.126 m; mean $|z|$ 6.074 $\rightarrow$ 1.027 m.

Final error: XY 9.843 m; $|z| = 3.601$ m.

Revisit Baselines

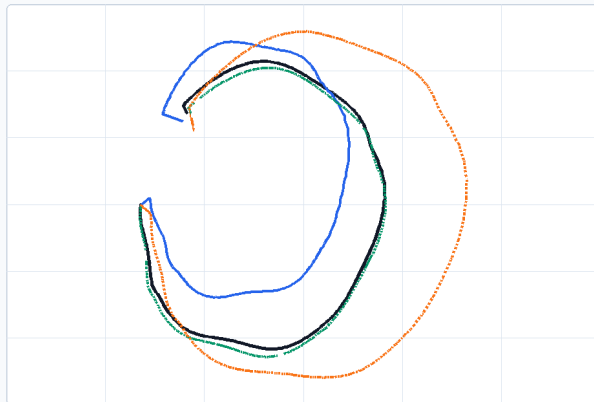
Pass A revisit is same-route; Pass B is harder because the route direction changes

Pass A and Pass A revisit realistic-drift baselines

Top-view uses the graph baseline. The z panel uses raw IMU/DVL odometry, where baseline spans 24.7 m and 24.8 m against GT spans 23.9 m and 23.9 m.

Pass A revisit is yaw-aligned to original Pass A for display only. Alignment yaw 70.1 deg.

top-view trajectory in online map frame



z (m)

2.9

z profile over normalised route progress

-4.4

-11.6

-18.9

-26.10



Revisit Result

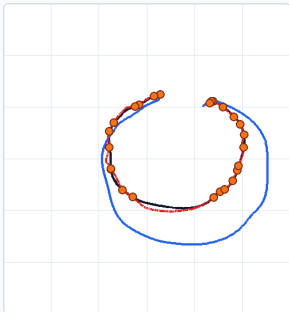
The revisit policy improves the hard case conservatively

Route-progress and same-route revisit replay results

Local single is strong for the same-route Pass A revisit. Route-progress gives guarded partial correction for the harder revisit cases.

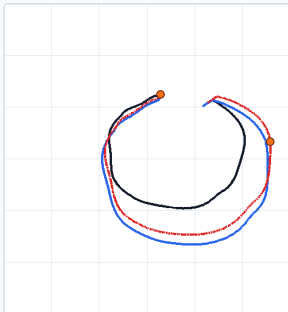
Pass A revisit local single

26 factors XY 2.901 to 1.008 m mean $|z|$ 12.123 to 1.034 m



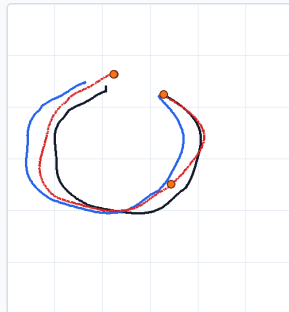
Pass A revisit route-progress

2 factors XY 2.901 to 2.442 m mean $|z|$ 12.123 to 1.946 m



Pass B route-progress

4 factors XY 2.734 to 1.875 m mean $|z|$ 13.115 to 3.605 m



— GT sonar mount

— baseline

- - - corrected

● accepted NAS-GS factor

Results Summary

The method works best as selective map-aided correction

Evidence

- Pass A shows strong same-route correction from a trained NAS-GS map.
- Pass A revisit confirms the local policy transfers to a newly recorded repeat route.
- Offshore BP replay transfers the same backbone to real sonar appearance.
- Accepted factors are interpretable because each one has a rendered-image explanation.

Boundary

- Reverse-route revisit improves on average, but endpoint stability is not solved.
- Wide search needs sequence confirmation before large corrections are trusted.
- Offshore accuracy is reference-consistency against Aqua-SLAM, not independent survey truth.

Future Work

The next steps follow directly from the remaining failure modes

Algorithm

- Strengthen sequence evidence for large-drift recovery.
- Improve route-progress endpoint stability for reverse revisits.

Validation

- Retrain or fine-tune NAS-GS with corrected trajectories.
- Repeat offshore tests with an independent reference if available.
- Move from replay to real robot timing, calibration, and safety tests.

Conclusion

NAS-GS can act as a useful sonar map prior for SLAM.

The implemented system turns render consistency into selective Pose3 registration factors and fuses them with IMU/DVL odometry through iSAM2.

It is strongest for same-route and repeated-inspection correction. Route-progress revisit and offshore replay show the method transfers, but large-drift recovery and independent real-world validation remain the main next steps.

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Thank you

Questions?